DoriaNET: A visual dataset from Hurricane Dorian for post-disaster building damage assessment

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Abstract

DoriaNET is a synthetic dataset for unmanned aerial vehicle (UAV)-based post-hurricane building damage assessment. DoriaNET consists of damage state annotations in accordance with FEMA Hurricane Model HAZUS-MH MR3 Manual. For each building, a damage state is manually assigned by closely matching the quantitative damage description of that building. Overall, DoriaNET provides the following information: (1) global coordinates (latitude and longitude) of each building appearing in the aerial footage; (2) local coordinates with pixel-level building boundaries (masks) in video frames; (3) wind-induced damage state for each building in accordance with FEMA Manual; (4) number of stories; and (5) an ordinal score representing damage annotation effort (easy, moderate, or difficult) for each labeled building.

Keywords: Hurricane Dorian, Post-disaster damage assessment, unmanned aerial vehicle (UAV), Visual dataset, Building damage.

Introduction

DoriaNET dataset aims to support research efforts on the emerging area of rapid and automated disaster damage assessment by creating and disseminating a structured visual dataset of post-disaster damage of buildings in accordance with the FEMA HAZUS Hurricane Model (FEMA, 2003). Specifically, DoriaNET consists of post-disaster aerial footage of Hurricane Dorian, a catastrophic Category 5 hurricane that made landfall in the Bahamas in August 2019. In our preliminary work (Cheng, Behzadan, & Noshadravan, 2021) that has led to the creation of DoriaNET, we have created a pilot dataset for building damage recorded in the aftermath of Hurricane Dorian. Particularly, we have utilized publicly available UAV footage (with creative commons licenses from YouTube) that captured a residential area in Marsh Harbor in the Bahamas. The DoriaNET dataset aims to scale up our previous effort by creating and publishing a more comprehensive visual dataset that can be of significant value to the natural hazards

research community, as well as motivate other researchers to invest their effort in dataset creation and sharing with the broader scientific community.

Dataset description

DoriaNET dataset contains the following information: (1) global coordinates (latitude and longitude) of each building in UAV footage; (2) local coordinates with pixel-level building boundaries (i.e., masks) in each video frame; (3) wind-induced damage state visually evaluated and assigned by the authors for each building in accordance with FEMA Hurricane Model HAZUS-MH MR3 Manual; (4) building characteristics (i.e., number of stories); and (5) an ordinal score representing damage annotation effort (0: easy, 1: moderate1, or 2: difficult) for each labeled building. To provide an organized and user-friendly dataset, the dataset is structured in a folder with individual video frames and corresponding mask images containing buildings local positions in each frame. Moreover, information about the disaster (e.g., location, time, intensity), and building objects (e.g., global coordinates, damage states, characteristics) are stored in a JSON file. Figure 1 shows a schematic visualization of DoriaNET structure.

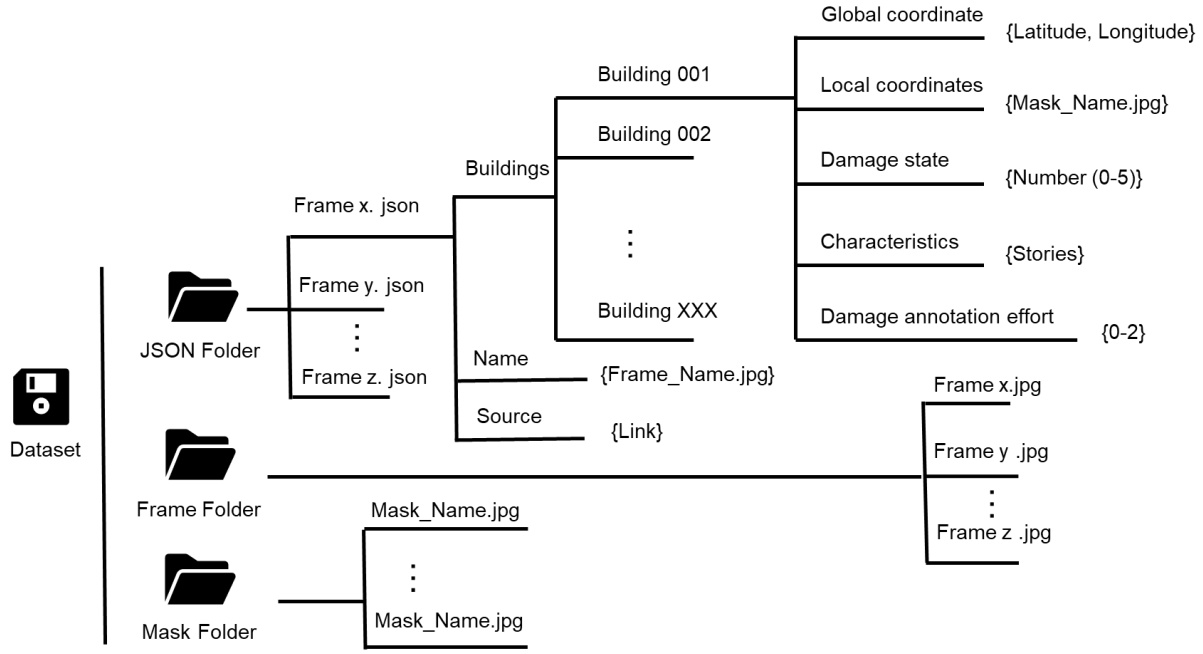

Figure 1. DoriaNET dataset structure

FEMA guidelines for wind-induced building damage present quantitative and straightforward definitions of five damage states ranging from 0 (least severe damage) to 4 (most severe damage), as shown in Table 1. Moreover, this damage rating system provides a detailed description of the criteria used to arrive at the final damage state by considering the extent of damage in key building components, including walls, roofs, decks, and windows. In DoriaNET, in addition to the standard damage states 0 through 4, we have introduced an additional category, state 5, to represent buildings under construction or destroyed (a category currently missing in the FEMA rating system) (Cheng et al., 2021). Overall, these six categories and their corresponding descriptions provide guidelines for rating building damage.

Table 1. Description of damage states in DoriaNET dataset (Cheng et al., 2021; FEMA, 2003)

Damage State	Quantitative Damage Description	Roof Cover Failure	Window Door Failures	Roof Deck	Missile Impacts on Walls	Roof Structure Failure	Wall Structure Failure
0	**No Damage or Very Minor Damage** Little or no visible damage from the outside. No broken windows, or failed roof deck. Minimal loss of roof over, with no or very limited water penetration.	<2%	No	No	No	No	No
1	**Minor Damage** Maximum of one broken window, door or garage door. Moderate roof cover loss that can be covered to prevent additional water entering the building. Marks or dents on walls requiring painting or patching for repair.	>2% and <15%	One window, door failure	No	<5 impacts	No	No
2	**Moderate Damage** Major roof cover damage, moderate window breakage. Minor roof sheathing failure. Some resulting damage to interior of building from water	>15% and <50%	>1% and the larger of 20% and 3	1 to 3 panels	5 to 10 impacts	No	No
3	**Severe Damage** Major window damage or roof sheathing loss. Major roof cover loss. Extensive damage to interior from water.	>50%	> the larger of 20% & 3 and <50%	>20 impacts	>20 impacts	No	No
4	**Destruction** Complete roof failure and/or, failure of wall frame. Loss of more than 50% of roof sheathing.	>50%	>50%	>25%	>20 impacts	Yes	Yes
5	**Buildings under Construction** Completed destroyed or under construction.	n.a.	n.a.	n.a.	n.a.	Yes	Yes

Imagery data of damaged buildings in DoriaNET were extracted from post-hurricane UAV videos obtained from YouTube. Web mining with keyword search was used to obtain these videos. Selected videos were manually inspected to ensure that they contain proper aerial views of disaster-affected buildings with varying degrees of building damage. Specifically, DoriaNET contains three 5-minute-long UAV videos recorded at 30 frames per second (FPS). These three videos have a 1280×720 pixel resolution and capture locations in Marsh Harbor (Lat 26°33'12.7"N; Long 77°05'44.5"W) in the Bahamas. All videos can be found using the links provided in the “related websites” section of this document. During pre-processing, to achieve a visually heterogeneous dataset, considering the frame rate of 30 FPS, one of every ten consecutive frames in each video are selected for annotation using Labelbox (Labelbox, 2019). An example frame is shown in Figure 2 (left). Additionally, frames that contain no building object (or buildings in a far distance from the camera viewpoint) are skipped. The annotation effort for each selected video frame involves drawing a mask along the boundaries of building objects, followed by assigning a numerical label that most closely represents the damage state of each individual building, as shown in Figure 2 (right).

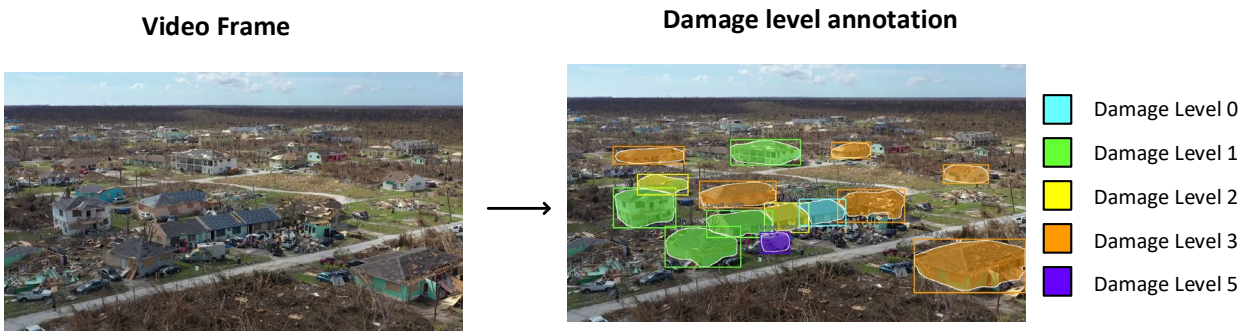

Figure 2. Visualization of the data annotation

As listed in Table 2, every instance (building object) in the DoriaNET dataset is assigned a single tag denoting the annotation effort (i.e., easy, moderate, or difficult) based on the amount of time and personal judgment of a human annotator for labeling. The "0: easy" tag represents samples where damage levels can be assigned with high confidence within 10 seconds. The "1: moderate" level denotes samples for which the labeler can decide a damage level with high confidence but need more than 10 seconds to annotate. Finally, "2: difficult" represents samples for which the labeler cannot decide the damage level (unsure between two or more levels), even when given ample time (more than 20 seconds). More details can be found in our previous work (Cheng et al., 2021).

Table 2 Damage level annotation effort (Cheng et al., 2021)

Annotation Effort	Description
0: Easy	Damage level is assigned with high confidence within 10 seconds.
1: Moderate	Damage level is assigned with high confidence, but may need more than 10 seconds of annotation time.
2: Difficult	Damage level is assigned with low confidence even when given ample time.

It should be noted that damage assessment can be challenging if the building is not fully visible in the camera view. For instance, if building components (e.g., walls, roof) are partially blocked by vegetation, smoke, haze, or other buildings, the annotator must decide based on what is visible, which may or may not represent the entire damage extent. However, when the same building is viewed from a different angle or height (as the UAV moves along its flight path), the damage annotation may be further revised. Keeping track of the global coordinates of each building, damage states in the dataset will be progressively updated. The global coordinates can be found in DoriaNET and are visualized in our shared DoriaNET Google Map shown in Figure 3. In total, DoriaNET contains 271 video frames which capture 147 buildings with 2,409 cropped images.

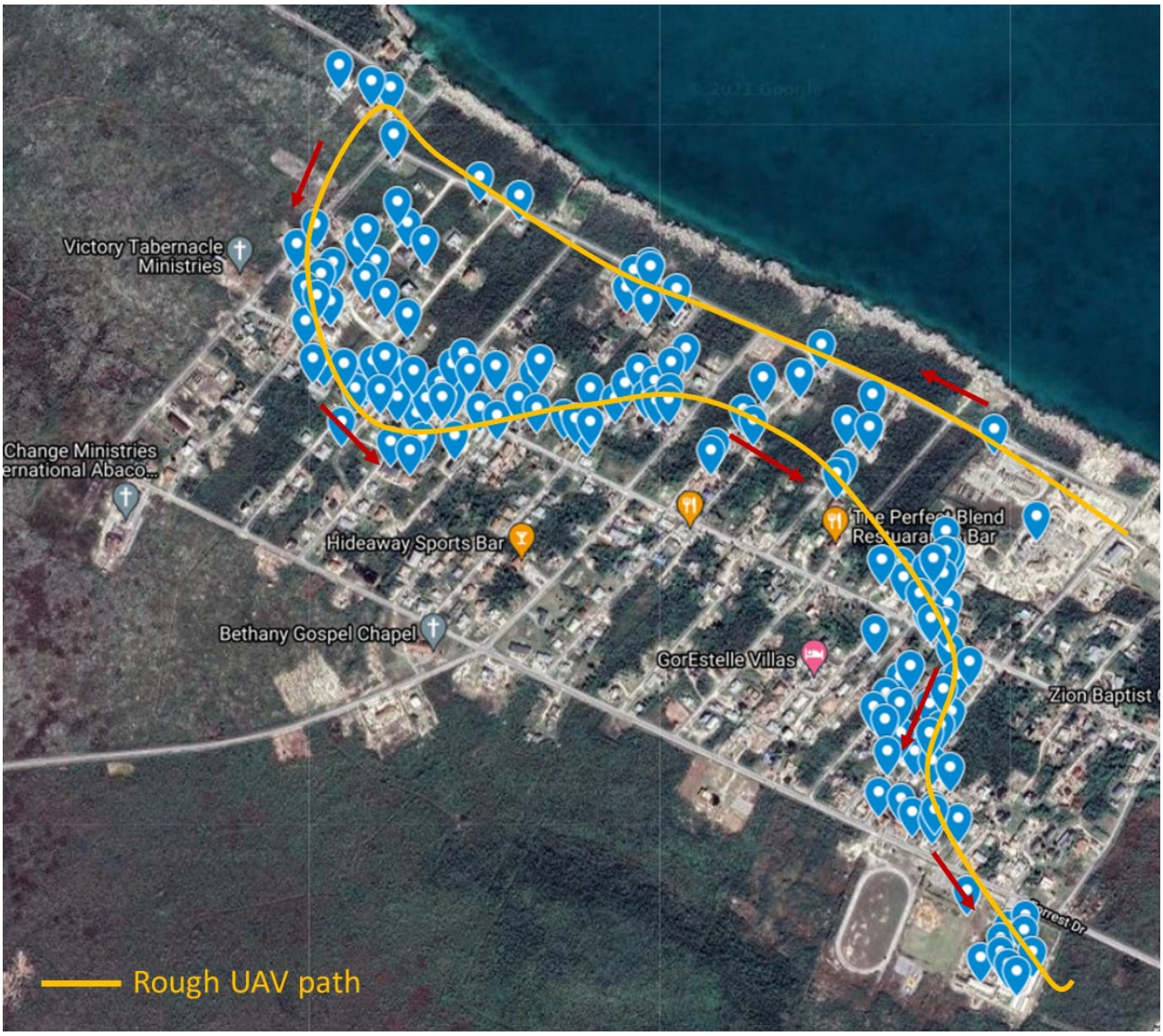

Figure 3. DoriaNET Google Map and an approximate UAV path

(Non-commercial use following Google Map Guideline: https://about.google/brand-resource-center/products-and-services/geo-guidelines)

In summary, DoriaNET is a first-of-a-kind visual dataset for hurricane damage assessment based on the FEMA HAZUS Hurricane Model, and can be used to train and test future visual recognition models (e.g., convolutional neural networks) to assist human inspection of damaged buildings and infrastructures.

Related websites

DoriaNET Google Map Link

Original UAV footages on YouTube (with creative common licenses):

- https://www.youtube.com/watch?v=T0GojxvrFm8
- https://www.youtube.com/watch?v=D8wvAgKNK5Q
- https://www.youtube.com/watch?v=Ya5dP0xqLyw

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